

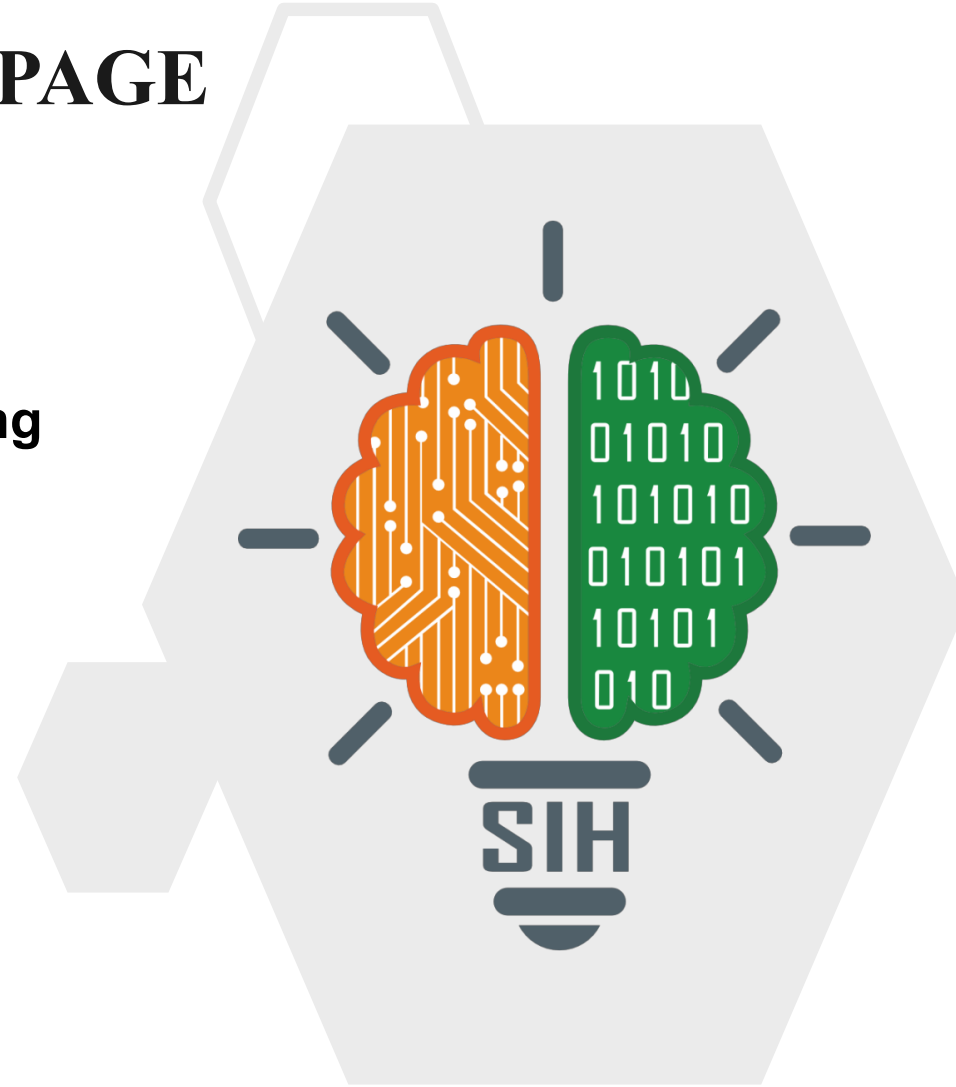
SMART INDIA HACKATHON 2026



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HACKATHON
2026

TITLE PAGE

- **Problem Statement ID – SIH26140**
- **Problem Statement Title – AI-Powered Interactive Platform for Learning, Designing and Simulating Quantum Algorithms**
- **Theme – Smart Education**
- **PS Category – Software**
- **Team ID – SIH-A0H-T004**
- **Team Name – ARISTELLA**

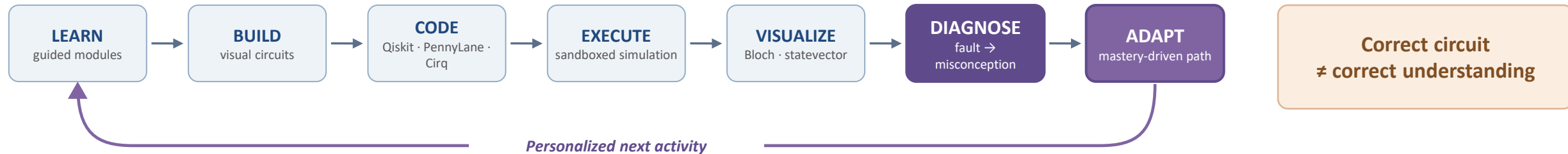


**Problem:**

- Quantum concepts like superposition, entanglement, phase, measurement are non-intuitive and non-visual.
- Existing resources are often static and theory-heavy; hardware access is scarce.
- Current tools primarily verify circuit correctness.
- A student can build a correct Bell state and still not understand WHY entanglement occurs.
- Procedural mastery $\neq$ conceptual mastery.

Our Idea:

- One integrated platform: Learn $\rightarrow$ Build $\rightarrow$ Code $\rightarrow$ Simulate $\rightarrow$ Visualize $\rightarrow$ Diagnose $\rightarrow$ Adapt.
- Every learner action captured as structured behavioural telemetry (gate placed / deleted, retry, failed run, hint requested).
- Deterministic diagnosis localizes the conceptual fault before the LLM generates an explanation.
- Behavioural evidence + conceptual assessment fused into per-concept mastery estimates.

**Proposed Solution — 8 Integrated Modules:**

01 Learning Modules Fundamentals, qubits, gates, entanglement, algorithms	02 Visual Circuit Builder Drag-drop gates, multi-qubit ops, live execution	03 Code Editor Qiskit / PennyLane / Cirq, extensible adapters	04 Simulation Engine Aer, PennyLane, Cirq; OpenQASM 3 as neutral IR
05 Visualization Bloch sphere, statevector, probabilities, step-through	06 AI Quantum Tutor Explanation, debugging, context-aware hints	07 Assessment Conceptual quizzes + circuit & coding challenges	08 Instructor Dashboard Concept mastery, misconception clusters, analytics

Innovation / Uniqueness:

- Calibration layer, not "platform + chatbot"** - the novelty is diagnosis, not the feature bundle
- Diagnosis before generation** - structural diff + step-wise state divergence locate the fault; the LLM only explains it
- Behaviour as evidence** - construction traces treated as indicators of conceptual state
- Procedural / conceptual divergence** - flags learners who build correctly but reason incorrectly
- Verified AI feedback** - every suggested fix is executed and checked before the student sees it

Technology Stack:

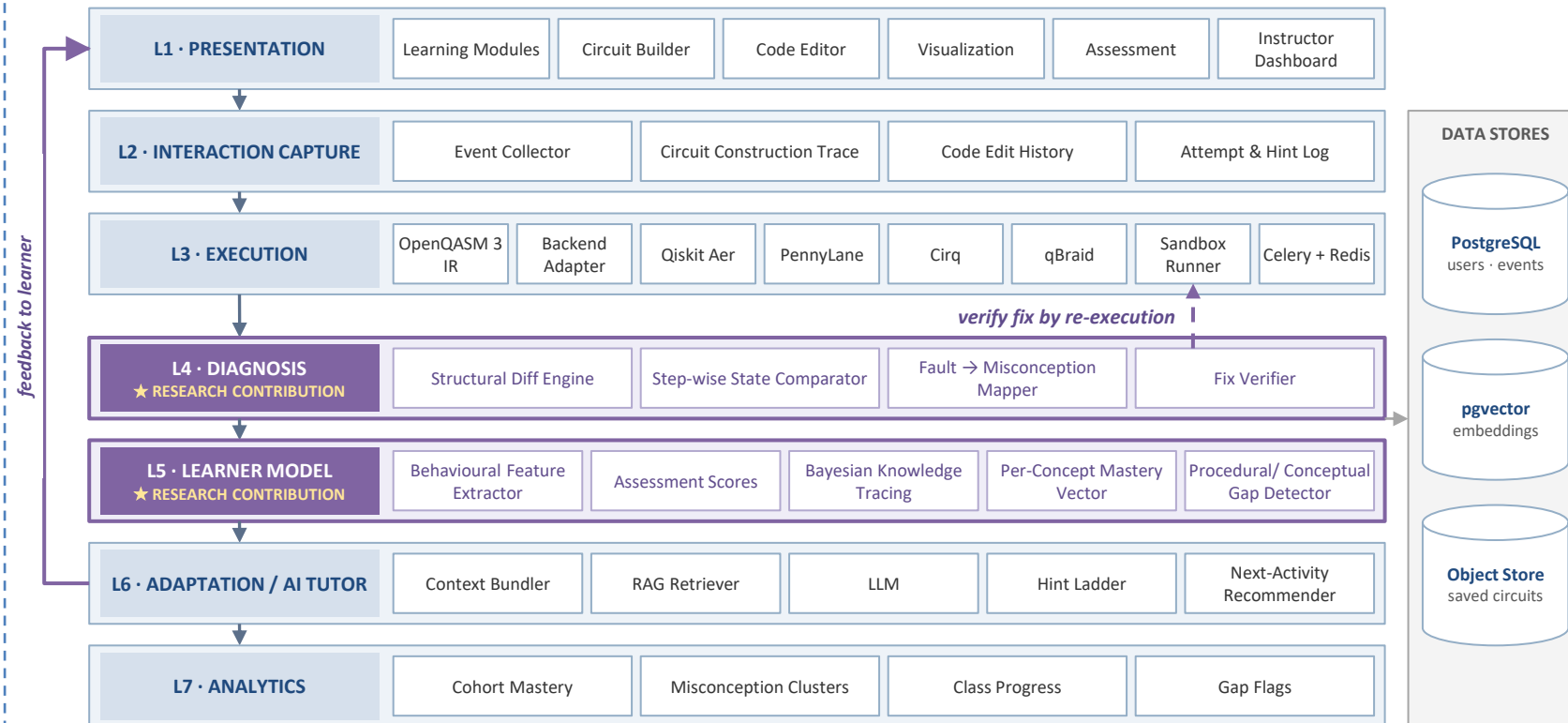
- **Frontend:** React + TypeScript, Vite, SVG/Canvas circuit editor, Monaco editor, Three.js (Bloch sphere), D3 / Plotly
- **Backend:** Python FastAPI, Celery + Redis job queue, sandboxed container execution
- **Quantum:** Qiskit Aer, PennyLane, Cirq; OpenQASM 3 as neutral IR; qBraid adapter
- **AI:** LLM API + RAG (pgvector); deterministic diagnosis engine upstream of the LLM
- **Learner Model:** scikit-learn, Bayesian Knowledge Tracing, logistic mastery estimation
- **Data:** PostgreSQL, pgvector, event-sourced telemetry
- **Infrastructure:** Docker, Kubernetes, WASM in-browser simulator

Scope:**MVP (SIH)**

- Grover + Bell-state fault catalogue
- 3 simulators
- Rule-based diagnosis
- BKT mastery

FUTURE

- Full algorithm coverage
- Learned diagnosis
- Real QPU
- LTI / LMS integration
- Collaborative workspaces

System Architecture:**End-to-End Workflow:**

Diagnosis: structural diff vs reference → step-wise state comparison (mod global phase) → fault-to-misconception label

Technical Feasibility:

01 Mature Quantum Tooling

Qiskit, PennyLane and Cirq provide mature, open-source quantum simulation.

Qiskit Aer

PennyLane

Cirq

02 Tractable Simulation

Classical simulation up to ~12 qubits is suitable for the MVP; OpenQASM 3 gives framework-neutral interchange.

≤ 12 qubits

OpenQASM 3

03 Deterministic Diagnosis

Diagnosis uses deterministic state comparison — no ML required for the MVP.

Structural diff

State comparison

04 Practical Web Stack

A standard, deployable stack for an interactive web platform.

React

FastAPI

PostgreSQL

Docker

Challenges:

01 Simulation cost / latency

02 LLM hallucination

03 Untrusted code execution

04 Sparse behavioural data

05 Behaviour-to-understanding validity

06 Limited real QPU access

07 Low-bandwidth institutions

Solutions:

12-qubit interactive cap · WASM for small circuits · caching · autoscaling workers

Deterministic diagnosis → RAG grounding → execute-and-verify every suggested fix

Sandboxed containers · CPU / memory / shot quotas · timeout controls

Rule-based misconception tagging + BKT + assessment signals

Treated as a testable hypothesis; validated through a pilot study

Noise-model simulation + limited cloud / QPU showcase

Offline WASM mode + version-pinned content

Scalability Roadmap:

PHASE 1 · SIH MVP

- Grover + Bell fault catalogue
- 3 backends
- Rule-based diagnosis
- Single classroom

PHASE 2 · PILOT

- Institution pilot
- Correlation study
- Dashboard hardening
- Full algorithm coverage

PHASE 3 · SCALE

- Multi-institution SaaS
- LTI / LMS integration
- Educator content-authoring tools

PHASE 4 · ECOSYSTEM

- Real QPU integration
- Open misconception item bank
- Regional-language support
- Open behavioural dataset



TARGET USERS & IMPACT

01 UG / PG STUDENTS

Hands-on quantum learning without hardware. Feedback on WHY a circuit is wrong.

02 INSTRUCTORS

Identify which concepts a cohort is struggling with; view misconception patterns and mastery.

03 INSTITUTIONS

Deployable quantum lab without hardware capex. Measurable learning outcomes.

04 SELF-LEARNERS

Structured beginner-to-advanced pathway with adaptive guidance.

05 EDUCATION RESEARCHERS

Structured learner-behaviour data for studying quantum learning.

BENEFIT ANALYSIS

01 Educational

- Detects procedural / conceptual gaps that correctness-only grading hides
- Diagnostic feedback names the misconception
- Adaptive sequencing driven by measured mastery

02 Social / Access

- Browser-based, no hardware requirement, offline-capable
- Suitable for institutions with limited access
- Supports instructors without formal quantum training

03 Economic / Strategic

- Reduces infrastructure requirements for quantum coursework
- Builds quantum-ready talent
- Behavioural-assessment methodology can extend to other abstract STEM domains

STUDENT ACTIVITY

BEHAVIOURAL DATA

CONCEPTUAL DIAGNOSIS

PERSONALIZED
INTERVENTIONMEASURED
UNDERSTANDING

INSTRUCTOR INSIGHT

QUANTUM-READY
WORKFORCE

Base Paper:

[Authors – verify], arXiv 2608.14459 - <https://arxiv.org/html/2608.14459>

Measuring student understanding in quantum computing: Development and validation of the Quantum Computing Conceptual Survey

Role: Validated conceptual-understanding instrument — the ground-truth measure for calibrating Qognita's behavioural signals.

Related Research:

01 — [Authors – verify], 2024 - <https://arxiv.org/abs/2405.19495>

Qiskit Code Assistant: Training LLMs for Generating Quantum Computing Code (arXiv 2405.19495)

→ Quantum code generation; explanation, repair & educational scaffolding not addressed

02 — [Authors – verify] - <https://arxiv.org/html/2211.16347>

Developing Programming Assignments for Teaching Quantum Computing and Quantum Programming (arXiv 2211.16347)

→ Statevector / unitary auto-assessment — correctness, not understanding

03 — [Authors – verify] - <https://arxiv.org/html/2604.06293>

The Quantum Education Ecosystem: A Review of Global Initiatives, Methods, and Challenges (arXiv 2604.06293)

→ Systematic evaluation of learning outcomes identified as lacking

04 — [Authors – verify] - <https://doi.org/10.1103/PhysRevSTPER.11.020135>

Student ability to distinguish between superposition states and mixed states in quantum mechanics — Phys. Rev. ST Phys. Educ. Res. 11, 020135

→ Documents a core misconception; not instrumented into a learning platform

05 — [Authors – verify] - <https://arxiv.org/html/2502.15709>

TutorLLM: Customizing Learning Recommendations with Knowledge Tracing and Retrieval-Augmented Generation (arXiv 2502.15709)

→ Knowledge tracing + RAG in generic domains; not applied to quantum circuits

Research Gap:

“Existing quantum-learning approaches provide interactive circuit construction, simulation, visualization, assessment and AI-based assistance, but there is limited work systematically correlating continuous learner behaviour during quantum circuit construction and problem-solving with validated measures of conceptual understanding.”

Tool-side work

→ Evaluates artifacts / correctness

Education-side work

→ Validates understanding, but limited observational access to construction behaviour

Knowledge tracing

→ Exists, but not over a quantum concept graph with circuits as the evidence stream

Our Research Contribution:**1 Behavioural Feature Schema**

Gate sequences, fault type, retries, hint depth — tagged to conceptual facets

2 Deterministic Diagnosis Layer

Structural diff + state divergence (mod global phase), upstream of the LLM

3 Learner Understanding Model

Fuses behaviour + assessment into per-concept mastery; flags divergence

4 Pilot Calibration Study

Mastery estimates vs a validated conceptual instrument; reported either way